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Experiment_6_Perceptron.m × +
C:\Users\Afroz\Desktop\NN\EXP 6\Experiment_6_Perceptron.m

1  clc;
2  clear;
3  close all;
4
5  %% Pattern Classification Using Perceptron Model
6
7  % Input patterns
8
9  % Class 1
10 X1 = [2 1;
11       3 2;
12       4 3;
13       5 3];
14
15 % Class 2
16 X2 = [1 4;
17       2 5;
18       3 6;
19       4 7];
20
21 % Combine input patterns
22 X = [X1; X2];
23
24 % Target outputs
25 % Class 1 = +1
26 % Class 2 = -1
27 T = [1; 1; 1; 1; -1; -1; -1; -1];
28
29 %% Add bias input
30 X_bias = [ones(size(X,1),1) X];
31
32 %% Initialize weights
33 rng(1); % For reproducible results
34 W = randn(1,3) * 0.1;
35
36 %% Learning parameters
37 learningRate = 0.1;
38 maxEpochs = 100;
39
40 %% Perceptron Training
41 for epoch = 1:maxEpochs
42     errorCount = 0;
43
44     for i = 1:size(X_bias,1)
45         % Calculate net input
46         net = W * X_bias(i,:);
47
48         % Perceptron activation function
49         if net >= 0
50             Y = 1;
51         else
52             Y = -1;
53         end
54
55         % Calculate error
56         error = T(i) - Y;
57     end
58 end
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59
60     % Update weights if classification is incorrect
61     if error ~= 0
62         W = W + learningRate * error * X_bias(i,:);
63         errorCount = errorCount + 1;
64     end
65
66 end
67
68 fprintf('Epoch %d: Misclassified Patterns = %d\n', ...
69         epoch, errorCount);
70
71 % Stop training if all patterns are correctly classified
72 if errorCount == 0
73     fprintf('\nTraining converged at Epoch %d\n', epoch);
74     break;
75 end
76
77 end
78

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79 %% Testing the trained perceptron

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80 Y_pred = zeros(size(T));
81
82 for i = 1:size(X_bias,1)
83
84     net = W * X_bias(i,:);
85
86     if net >= 0
87         Y_pred(i) = 1;
88     else
89         Y_pred(i) = -1;
90     end
91
92 end
93

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94 %% Display Results

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95 fprintf('\nPattern Classification Results\n');
96 fprintf('-----\n');
97 fprintf('X1\tX2\tTarget\tPredicted\tStatus\n');
98 fprintf('-----\n');
99
100 for i = 1:size(X,1)
101
102     if T(i) == Y_pred(i)
103         status = 'Correct';
104     else
105         status = 'Wrong';
106     end
107
108     fprintf('%d\t%d\t%d\t%d\t\t%s\n', ...
109             X(i,1), X(i,2), T(i), Y_pred(i), status);
110
111 end
112

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113 %% Calculate Accuracy
114 accuracy = sum(Y_pred == T) / length(T) * 100;
115
116 fprintf('\nClassification Accuracy = %.2f%%\n', accuracy);
117

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118 %% Display Final Weights
119 fprintf('\nFinal Weights:\n');
120 fprintf('Bias Weight (W0) = %.4f\n', W(1));
121 fprintf('Weight W1 = %.4f\n', W(2));
122 fprintf('Weight W2 = %.4f\n', W(3));
123

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124 %% Plot the patterns
125 figure;
126 hold on;
127 grid on;
128
129 % Plot Class 1
130 plot(X1(:,1), X1(:,2), 'ro', ...
131      'MarkerSize', 8, 'LineWidth', 2);
132
133 % Plot Class 2
134 plot(X2(:,1), X2(:,2), 'bs', ...
135      'MarkerSize', 8, 'LineWidth', 2);
136

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137 %% Plot decision boundary
138 x1 = linspace(0,6,100);
139
140 % Equation:
141  $W_0 + W_1 \cdot x_1 + W_2 \cdot x_2 = 0$ 
142  $x_2 = -(W(1) + W(2) \cdot x_1) / W(3);$ 
143
144 plot(x1, x2, 'k-', 'LineWidth', 2);
145
146 xlabel('Feature X1');
147 ylabel('Feature X2');
148 title('Pattern Classification Using Perceptron');
149
150 legend('Class 1', 'Class 2', 'Decision Boundary', ...
151      'Location', 'best');
152
153 hold off;

```

Epoch 1: Misclassified Patterns = 0

Training converged at Epoch 1

Pattern Classification Results

X1	X2	Target	Predicted	Status
2	1	1	1	Correct
3	2	1	1	Correct
4	3	1	1	Correct
5	3	1	1	Correct
1	4	-1	-1	Correct
2	5	-1	-1	Correct
3	6	-1	-1	Correct
4	7	-1	-1	Correct

Classification Accuracy = 100.00%

Final Weights:

Bias Weight (W_0) = -0.0649

Weight W_1 = 0.1181

Weight W_2 = -0.0758

